

Gabriel Virto Chirhuana

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https://bit.ly/Gabriel_Virto_Portfolio

Education

Technical University of Munich, Munich, Germany

Apr 2025 - Present

M.Sc. Aerospace Engineering

Technical University of Munich, Munich, Germany

Graduated: 2025

B.Sc. Mechanical Engineering

Thesis: *Modeling of Convective Heat Transfer in Automotive Headlamps for Simulation-Based Design*

Engineering Tools

Programming: MATLAB/Simulink, Python, Modelica (Dymola)

CFD: ANSYS CFX, ANSYS Fluent, Simcenter STAR-CCM+

FEA: Nastran Hypermesh/Optistruct

CAD: CATIA V5, SolidWorks

Professional Experience

BMW Group, Munich, Germany

Working Student – Simulation Engineering in R&D Power Systems

Jun 2024 – May 2026

- Developed and refined system-level thermal-electrical models in Modelica (Dymola) to simulate power system behavior under dynamic operating conditions
- Built and integrated multi-domain component models, enabling analysis of system interactions and failure scenarios in vehicle energy networks
- Analyzed measurement data (MATLAB, Excel) to characterize thermal and electrical behavior of power electronic components and support model development
- Performed model calibration and parameter optimization against experimental data, improving simulation fidelity and reducing deviation between test and model results
- Generated model-based training datasets for data-driven system identification using neural ODE approaches (Simulink)
- Conducted technical research and structured analyses on energy network simulation topics, supporting advanced modeling and simulation workflows

FORVIA HELLA, Lippstadt, Germany

Intern – Thermal Engineering in R&D Automotive Headlamps

Aug 2023 – May 2024

- Advanced a Modelica-based simulation model for automotive headlamp design, improving system-level thermal analysis capabilities
- Extended the model to capture convective heat transfer and internal flow phenomena, increasing physical accuracy of thermal predictions
- Expanded and structured a Modelica library, enabling modular and scalable simulation workflows
- Validated simulation results against CFD analyses, ensuring consistency between reduced-order models and high-fidelity simulations
- Assessed model transferability across different headlamp architectures, supporting reuse and standardization of simulation approaches
- Conducted systematic parameter studies and sensitivity analyses to evaluate thermal behavior and identify key design drivers

Metallbau & Design – Brüll + Gruber GmbH, Munich, Germany

Intern – Manufacturing Processes

Jul 2020 – Sep 2020

- Operated manual and CNC machining tools, gaining hands-on experience in manufacturing processes
- Applied knowledge of tolerances, fabrication limits, and design constraints in real-world production
- Strengthened understanding of design-for-manufacturing principles

Aerospace Projects

CAD in Aircraft Design – Beechcraft B58 Model 2026

Modeled a full aircraft assembly in Catia V5, including fuselage, empennage, nacelle, and propeller, and produced detailed engineering drawings.

CFD in Turbomachinery Design – Axial Compressor Analysis 2026

Performed 2D and 3D CFD simulations in Ansys CFX to evaluate aerodynamic performance and blade loading of an axial compressor stage. Conducted a comparative study on reduced stator blade count and its impact on flow behavior and efficiency.

Space Mission Design – Lunar IoT Constellation 2025

Developed thermal control strategies for a lunar satellite constellation by analyzing thermal cycles, eclipse periods, and power constraints, maintaining system temperatures within operational limits (target: $< 35^{\circ}\text{C}$).

Aircraft Aerodynamics – Wind Tunnel Experiment for Static Stability 2025

Conducted wind tunnel experiments using pressure and force measurements to assess aerodynamic performance and static stability with and without a horizontal stabilizer.

Material Selection for Additive Manufacturing of a Turbine Blade 2022

Evaluated candidate materials for a high-pressure turbine blade under combined thermal and mechanical loads. Designed blade geometry in SolidWorks and analyzed flow behavior using Ansys Fluent, comparing performance, manufacturability, and cost to identify optimal design trade-offs.

Student Engineering Teams

Impetus – Sailing Team, Munich, Germany

Manufacturing Team Member

Apr 2025 – Sep 2025

- Helped build a student-designed high-performance sailing boat
- Worked with lightweight structural materials during the manufacturing process

Phantum Student Club, Munich, Germany

Propulsion Team Member

May 2024 – Dec 2024

- Part of the propulsion team developing a small high-performance UAV
- Worked on the conceptual design of an afterburner for small turbojet engines
- Designed a water-injection system in Solidworks aimed at thrust augmentation and temperature control

Languages

Spanish – Native | German – C1 | English – C1 | Italian – A2